

	Case Name: MEJORAMIENTO DEL PARQUE Project Owner: Eder Omar Vilca Carrillo	Sector	Construction	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
User defined distributions				
Submitted by	Matilde Casado and Margarida Antunes			
Date	2025	Project Control	Automatic tracking	
File Name	C2025-26		Tracking based on user input	

1. Project description

Project authenticity

This project upgrades San Salvador's main park with better infrastructure, lighting, green areas, and drainage to enhance community use.
The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	85
Planned Duration (PD)	120 days
Budget At Completion (BAC)	1 249 157.0€
Renewable Resources	9
Consumable Resources	-

* standard eight-hour 7 days of the week

Network topology	
Serial/Parallel (SP)	30%
Activity Distribution (AD)	32%
Length of Arcs (LA)	0%
Topological Float (TF)	42%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	0	0	N/A
CRI-tau	0	0	N/A

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	0	0	N/A
CRI-rho	0	0	N/A
CRI-tau	0	0	N/A

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	7.1	25.8	3.4
SI	9.2	25.7	3.2
SSI	2.8	11.5	4.1
CRI-r	9.6	10.2	2.2
CRI-rho	9.5	10.0	2.0
CRI-tau	8.8	7.3	1.3

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	10.7	-2.6	1	0.0	0.0
PV - SPI	11.4	-2.8	CPI	0.0	0.0
PV - SCI	11.4	-2.8	SPI	1.0	-0.2
ED - 1	10.2	-3.0	SPI(t)	0.9	-0.3
ED - SPI	11.1	-3.1	SCI	1.0	-0.2
ED - SCI	11.1	-3.1	SCI(t)	0.9	-0.3
ES - 1	11.0	-7.1	0.8 CPI + 0.2 SPI	0.2	0.0
ES - SPI(t)	16.9	-11.7	0.8 CPI + 0.2 SPI(t)	0.2	0.0
ES - SCI(t)	16.9	-11.7			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 1 tracking period. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	0.0	680062.3	1587.0	1.0	7.5	18.2	0.8
std dev	N/A	641367.5	1927.6	0.0	4.7	17.8	0.2
final	0.0	1133577.6	2950.0	1.0	10.8	30.8	1.0

2.3.3.2. Time forecasting

PD	120 days	Real Duration	12 days	Early	-90%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	409.5	0.7	1656.3	-1656.3
PV - SPI	410.5	0.7	1660.4	-1660.4
PV - SCI	411.5	0.7	1664.6	-1664.6
ED - 1	412.5	0.7	1668.7	-1668.8
ED - SPI	413.5	0.7	672.9	-672.9
ED - SCI	414.5	0.7	1677.1	-1677.1
ES - 1	415.5	0.7	1681.3	-1681.3
ES - SPI(t)	416.5	0.7	1685.4	-1685.4
ES - SCI(t)	417.5	0.7	1689.6	-1689.6

2.3.3.3. Cost forecasting

BAC	1 249 157.0€	Real Cost	1 249 157.0€	On Budget	0%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	1 249 157.0	0.0	0.0	0.0
CPI	1 249 157.0	0.0	0.0	0.0
SPI	888 757.2	509682.8	14.4	14.4
SPI(t)	859 234.9	551433.7	15.6	15.6
SCI	888 757.2	509682.8	14.4	14.4
SCI(t)	859 234.9	551433.7	15.6	15.6
0.8 CPI + 0.2 SPI	1 065 882.3	259190.1	7.3	7.3
0.8 CPI + 0.2 SPI(t)	1 022 187.6	320983.8	9.1	9.1